

COMPUTER NETWORKS

Computer Networks: Architecture, Protocols & Systems

From OSI & TCP/IP Layering to High-Performance Routing, Congestion Control & Network Security

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CHAPTER 1

Network Topologies, Layering Models & Physical Transmission

1. Theoretical Foundations & Scope

Computer networking provides the communication backbone of modern computing. This chapter covers the OSI 7-layer reference model, TCP/IP protocol suite, transmission media, packet switching versus circuit switching, and framing techniques.

2. Core Conceptual Framework

- **OSI vs TCP/IP:** Comparing theoretical 7-layer encapsulation with practical 4-layer TCP/IP internet protocol stack architecture.
- **Multiplexing & Framing:** FDM, TDM, and WDM methods alongside bit-stuffing and byte-stuffing error detection mechanisms.
- **Data Link Layer Control:** Flow control protocols including Stop-and-Wait, Go-Back-N, and Selective Repeat ARQ mechanisms.

Key Practical Insights & Examination Takeaways:

- ' Understand layering abstractions and encapsulation headers.
- ' Analyze channel capacity using Nyquist and Shannon theorems.
- ' Differentiate framing algorithms and error detection codes like CRC.

CHAPTER 2

Network Layer Routing Algorithms & IP Addressing

1. Theoretical Foundations & Scope

The network layer is responsible for host-to-host packet delivery. We examine IPv4 and IPv6 addressing schemes, CIDR subnetting, distance-vector algorithms (Bellman-Ford), link-state routing (Dijkstra), and BGP peering.

2. Core Conceptual Framework

- **Dijkstra vs Bellman-Ford:** Link-state global knowledge calculation versus distance-vector distributed routing updates and count-to-infinity mitigation.
- **CIDR & Subnetting:** Variable-length subnet masking (VLSM) and hierarchical aggregation for efficient routing tables.
- **Network Address Translation:** NAT and PAT translation mechanics allowing private subnets to share public IP addresses.

Key Practical Insights & Examination Takeaways:

- ' Calculate subnet masks, usable host ranges, and broadcast addresses.
- ' Trace shortest path algorithms across weighted network graphs.
- ' Contrast interior gateway protocols (OSPF, RIP) with exterior protocols (BGP).

CHAPTER 3

Transport Layer Protocols: TCP, UDP & Congestion Control

1. Theoretical Foundations & Scope

Transport layer protocols provide process-to-process communication. We analyze TCP three-way handshakes, connection termination, window-based flow control, and AIMD congestion control mechanics.

2. Core Conceptual Framework

- **TCP vs UDP:** Connection-oriented reliable byte-stream transmission versus low-overhead connectionless datagram streaming.
- **Congestion Control:** Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery algorithms.
- **Port Multiplexing:** Socket pair abstractions binding local IP/port combinations to remote IP/port endpoints.

Key Practical Insights & Examination Takeaways:

- ' Examine sequence numbers, acknowledgment numbers, and TCP state transitions.
- ' Explain why UDP is preferred for real-time gaming and VoIP streaming.
- ' Analyze bufferbloat and active queue management (RED).

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

